

Day 3 Write-Up: Feature Engineering & Cross-Validated Model Comparison

Adult Income Dataset

1. Engineered Feature List

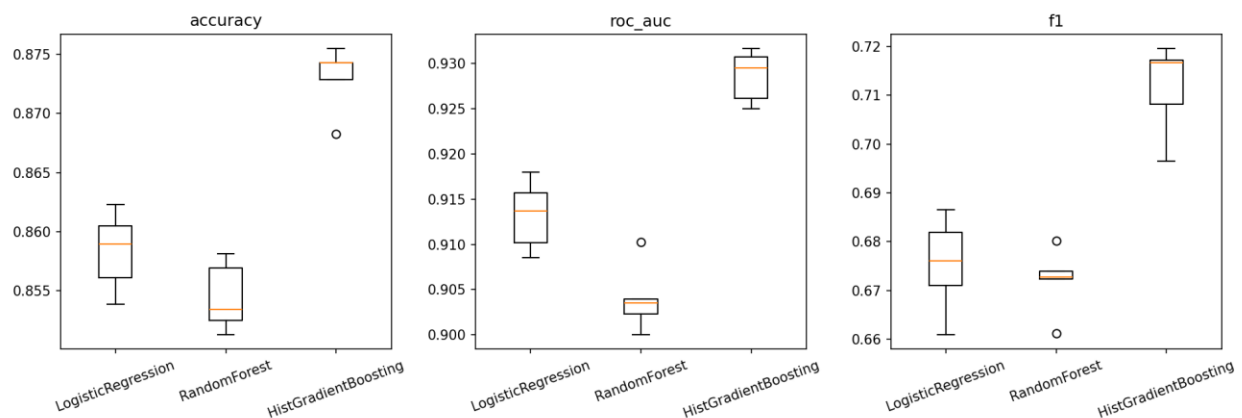
Feature	Type	Rule	Signal (mutual info)
log_capital_gain	numeric	$\log_{1p}(\text{capital-gain})$	0.084 (highest)
edu_hours_interaction	numeric	$\text{education-num} \times \text{hours-per-week}$	0.083
age_bucket	categorical	age binned into 6 groups	0.061
higher_ed	boolean	$\text{education-num} \geq 13$	0.048
hours_bin	categorical	hours binned: part/full/overtime	0.037
has_capital_gain	boolean	$\text{capital-gain} > 0$	0.030

log_capital_gain and edu_hours_interaction carry the strongest univariate signal of the six. Grouped target rates confirm the intuition behind the two binned features: >50K rate rises from 1.1% (<25) to a peak of 39.4% (45-54) then falls for 65+, and rises steadily from 8.4% (part-time) to 40.0% (overtime).

2. Cross-Validated Model Comparison

StratifiedKfold (k=5) was used to compare Logistic Regression, Random Forest, and HistGradientBoosting, all sharing the identical feature-engineering + preprocessing pipeline. Primary metric: accuracy.

Model	Accuracy (mean±std)	ROC AUC (mean±std)	F1 (mean±std)
Logistic Regression	0.858 ± 0.003	0.913 ± 0.004	0.675 ± 0.009
Random Forest	0.854 ± 0.003	0.904 ± 0.003	0.672 ± 0.006
HistGradientBoosting	0.873 ± 0.003	0.929 ± 0.003	0.712 ± 0.009



HistGradientBoosting leads on all three metrics with the tightest spread across folds, confirming it as the strongest of the three on this engineered feature set.

3. Statistical Test Results

Paired comparison (same 5 folds) between the top 2 models by mean accuracy: HistGradientBoosting (0.873) vs. Logistic Regression (0.858).

Test	Statistic	p-value
Paired t-test	$t = 16.42$	$p \approx 0.00008$
Wilcoxon signed-rank	$W = 0.0$	$p \approx 0.0625$

HistGradientBoosting beats Logistic Regression on every single one of the 5 folds, and the t-test p-value is far below 0.05 — the gap is very unlikely to be random noise. The Wilcoxon p-value (≈ 0.06) sits just above the usual 0.05 cutoff, which is expected: with only 5 paired samples, Wilcoxon has very little statistical power regardless of how consistent the pattern is. Practically, both tests agree on the direction and consistency of the difference, so the result is treated as a real (if modest, ~ 1.5 point) improvement, not a precise p-value to over-interpret.

4. Feature Importance (Engineered Features)

Random Forest (tree-based): `edu_hours_interaction` ranks #3 out of 118 total post-encoding features (importance ≈ 0.075), just behind `fnlwgt` and `age`, and ahead of every one-hot category. `log_capital_gain` is also top-10 (≈ 0.044). The boolean flags and binned features rank lower individually, since they're coarser versions of signal the raw numeric columns and the interaction term already capture.

Logistic Regression (coefficients): `log_capital_gain` (+4.15) and `has_capital_gain` (−4.15, opposite sign because it's the raw flag being “corrected” by the log version) are by far its largest-magnitude engineered coefficients, followed by the `age_bucket` bins (65+ and <25 both negative, mid-career bins positive).

Both findings make sense: `edu_hours_interaction` encodes “highly educated and working full-time,” a strong real income signal a linear model or tree can't get from the two source columns separately, and the log transform is exactly what a linear model needed to use the skewed capital-gain signal well.

5. Recommended Models & Features for Tuning

Feature selection check: `SelectKBest(mutual_info_classif, k=40)` was compared against the full ~ 118 -column feature set using HistGradientBoosting. Selection reduced accuracy (0.869 vs 0.873), ROC AUC (0.926 vs 0.929), and F1 (0.698 vs 0.712), and took longer per fold (mutual information must be recomputed on every training fold).

Feature set	Accuracy	ROC AUC	F1	CV time
Full (~118 cols)	0.873	0.929	0.712	15.7s
SelectKBest k=40	0.869	0.926	0.698	87.6s

- Model to tune tomorrow: HistGradientBoosting — best on accuracy, ROC AUC, and F1, with the tightest fold-to-fold spread, and the lead over Logistic Regression is statistically consistent (wins all 5 folds).
- Feature set to keep: the full engineered feature set (original columns + all 6 new features). SelectKBest with mutual information didn't help and cost more time, so no columns are dropped going into tuning.
- Logistic Regression stays as the interpretable reference model, since its coefficients are still useful for explaining predictions even though it's not the top performer.